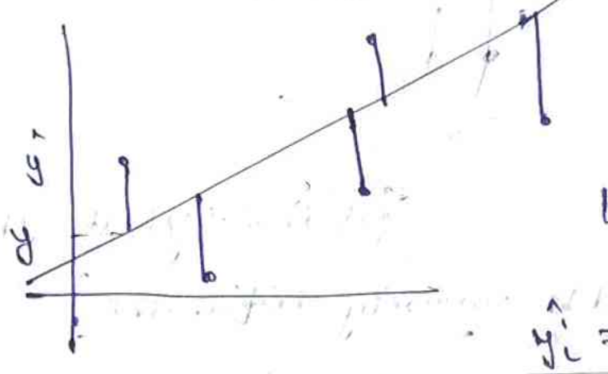
Gradient descent: A first order iterative optimization ~~technique~~ algorithm used to find a local minimum of a differentiable function.



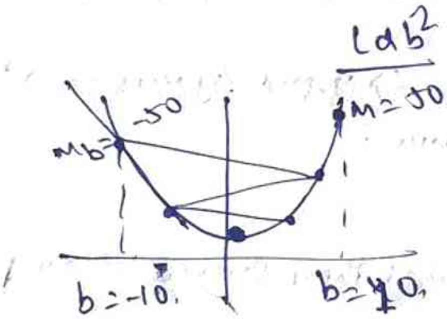
$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\hat{y}_i = mx_i + b$$

$$\boxed{L = \sum_{i=1}^{n} (y_i - mx_i - b)^2} \quad L(m,b)$$

Assume $m = 78.35$

$$\left(L = \sum_{i=1}^{n} (y_i - 78.35x_i - b)^2 \right) \quad L(b)$$



S1 → Assume a random value of b.

S2 → $\boxed{b_{new} = b_{old} - m_b}$

C1 → $b = -10$

$b_n = -10 - (-50)$

$= -10 + 50 = 40$

if $b \to$ (-ve) (Increment)

$b \to$ +ve (Decrement)

C2: $b = 10$.

$b_n = 10 - (50)$

$= -40$.

~~keep repeating until~~ :

* ~~no steps~~ As there are many drastic changes, we may miss the minima, To overcome this,

$\boxed{b_{new} = b_{old} - \eta \text{ slope}}$ $\quad \eta \to$ learning rate

$\boxed{\eta = 0.01}$

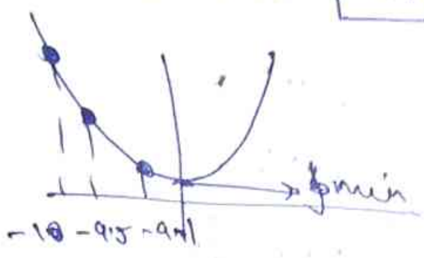
$b_{new} = -10 - (0.01 \times -50)$

$= -10 + 0.5 = \boxed{-9.5}$

Assume $m_{-9.5} = -40$

$b_{new} = -9.5 - (0.01 \times -40)$

$= -9.5 + 0.4 = \boxed{-9.1}$ (small changes approaching minima.



$\boxed{b_{new} = b_{old} - \eta m}$ → step size.

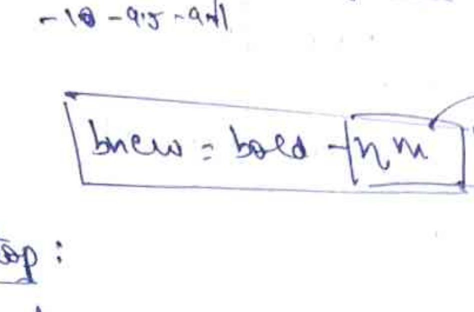
When to stop:

1. $b_{new} - b_{old} = 0.0001$ (converge)

2. We limit the no of iterations
$\boxed{n = 1000}$ (Depends on scenario)
epochs

Mathematical formulae:

$m = 78.35$



Step 1: Start with a random value of b.

for i in epochs: → 1000

$b_{new} = b_{old} - \eta \cdot slope$ → 0.01

Assume $b = 0$,

$m_{b=0} = ?$

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$m_{b=0} = \frac{dL}{dB} = \sum \frac{d}{dB} (y_i - mx_i - b)^2$$

$$= \sum 2 (y_i - mx_i - b)(-1) \quad b = 0$$

$$m = -2 \left(\sum_{i=1}^{n} (y_i - 78.35 \times x_i) \right)$$

$$m_b = 0 = -2\sum_{i=1}^{n}(y_i - 78.35x_i)$$

$$b_n = b_0 - 0.01 m_b$$

Effect of m:

$$m = 1, b = 0$$

$$\boxed{\begin{aligned} b &= b - \eta\, slope_{b=0} \\ m &= m - \eta\, slope_{m=1} \end{aligned}}$$

$$L = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$L = \sum_{i=1}^{n}(y_i - mx_i - b)$$

$$\begin{cases} slope_{b=0} = \left.\dfrac{\partial L}{\partial b}\right|_{b=0} \\ slope_{m=1} \left.\dfrac{\partial L}{\partial m}\right|_{m=1} \end{cases}$$

Types of GD:

1) Batch GD
2) Stocastic GD
3) Mini Batch GD.

$m = 1 \quad b = 0$

iterate $\longrightarrow \dfrac{\partial L}{\partial m}$.

$$m = m_0 - n \times slope|_{m=1}$$

$$b = b_0 - n \cdot slope|_{b=0} \longrightarrow \frac{\partial L}{\partial b}$$

when value of m is updated after taking data of all rows $\longrightarrow$ Batch. (Small DS)

When value updated by each row

$\longrightarrow$ Stocastic

mini batch ⟶ define a batch size (30), for total 1000 rows, 10 times m, b updated

Batch Gradient Descent:

Mathematical formulation:

cgpa	iq	lpa
x_1	x_2	y (lpa)
8.1	9.3	3.2
7.5	9.5	3.5

$$[Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2]$$

$$\{\beta_0, \beta_1, \beta_2\}$$

1. Random value.

$$\beta_0 = 0, \quad \beta_1, \beta_2 = 1$$

2. Apply GD:

epochs = 100, η = 0.1

$L(\beta_0, \beta_1, \beta_2) \rightarrow$ GD

$$\beta_0 = \beta_0 - \eta\,\text{slope} \rightarrow \frac{\partial L}{\partial \beta_0}$$

$$\beta_1 = \beta_1 - \eta\,\text{slope} \rightarrow \frac{\partial L}{\partial \beta_1}$$

$$\beta_2 = \beta_2 - \eta\,\text{slope} \rightarrow \frac{\partial L}{\partial \beta_2}$$

MSE

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

{2 rows, 3 cols}
$n=2$

$$= \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$= \frac{1}{2}\left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2\right]$$

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12}$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22}$$

$$L = \frac{1}{2}\left[(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2\right]$$

$$\frac{\partial L}{\partial \beta_0} = \frac{1}{2}\left(2(y_1 - \hat{y}_1)(-1) + 2(y_2 - \hat{y}_2)(-1)\right)$$

Assume n rows

$$-\frac{2}{n}\left[(y_1-\hat{y}_1)+(y_2-\hat{y}_2)\dots\dots\quad y_n-\hat{y}_n)\right]$$

$$\boxed{\frac{\partial L}{\partial \beta_0}=\left[-\frac{2}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)\right]}$$

$$\frac{\partial L}{\partial \beta_1}=-\frac{2}{2}\left[(y_1-\hat{y}_1)(x_{11})+(y_2-\hat{y}_2)(x_{21})\right]$$

for n rows,

$$\boxed{\frac{\partial L}{\partial \beta_1}=-\frac{2}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)(x_{i1})}$$

$$\boxed{\frac{\partial L}{\partial \beta_2}=-\frac{2}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)(x_{i2})}$$

x_{i1} represents data of n rows and 1st colmn (x_1)

n rows and 2nd colm (x_2)